Then, by applying the Theorem-8.5 for simultaneous occurrence of these four events, it follows that

$$P(B) = \frac{6}{19} \times \frac{5}{18} \times \frac{4}{17} \times \frac{3}{16}$$

$$P(B) = \frac{5}{1292}.$$

(ii) Suppose that

C: In the randomly selected students the first two are boys and the next are girls

There will be 13 boys among the 19 students, in total, while selecting the first student; there will be 12 boys among 18 students, in total, while selecting the second student; 6 girls among 17 students, in total, while selecting the third student; and 5 girls among the remaining 16 students, in total, while selecting the fourth student.

Then, by applying the Theorem 8.5 for simultaneous occurrence of these four events, it follows that

$$P(C) = \frac{13}{19} \times \frac{12}{18} \times \frac{6}{17} \times \frac{5}{16} = \frac{65}{1292}$$

8.8 Bayes' Theorem and its Applications

In some cases, probability for the occurrence of an event of interest A may be difficult to compute from the given information. But, it may be possible to calculate its conditional probabilities $P(A/B)$ and $P(A/\bar{B})$ for some other event B of the same experiment. Then, $P(A)$ can be calculated applying the law of total probability. This theorem is a prelude for Bayes' theorem.

Theorem 8.6 (Law of Total Probability)

If $B_1, B_2, \ldots, B_n$ are mutually exclusive events such that $\bigcup_{j=1}^{n} B_j = S$ and $P(B_j) > 0$ for $j = 1, 2, \ldots, n$, Then for any event A

$$P(A) = P(A/B_1)P(B_1) + P(A/B_2)P(B_2) + \ldots + P(A/B_n)P(B_n).$$

In real life situations, decision making is an ongoing process. Situations may arise where we are interested in an event on an ongoing basis. Every time some new information may be available and based on this the probability of the event should be revised. This revision of probability with additional information is formalized in probability theory in the theorem known as Bayes' Theorem.